MentalHospital: A Virtual Environment for Evaluating Psychiatric Clinical Encounters

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Abstract

Large language models (LLMs) have shown strong performance on isolated psychiatric tasks, including dialogue, diagnosis, and treatment planning, yet existing benchmarks rarely simulate complete psychiatric clinical encounters. We introduce **MentalHospital**, a virtual evaluation environment for LLM-based psychiatric clinical encounters. MentalHospital instantiates the Subjective Interviewing, Objective Examination, Diagnostic Assessment, and Treatment Planning (S.O.A.P.) workflow, using skill-augmented standardized patients constructed from 1,193 de-identified psychiatric electronic health record (EHR) cases spanning all major ICD-11 categories and 76 disorders. Each encounter is assessed through a dual-track protocol that combines objective comparison against EHR-derived references with subjective assessment of clinical process quality. To scale specialist judgment, we develop **MentalEval**, five domain-specific evaluators covering communication empathy, interviewing professionalism, clinical-note quality, diagnostic rigor, and treatment appropriateness, trained with rubric-grounded SFT and expert-guided DPO. Survey responses from 22 clinicians support MentalHospital's clinical fidelity (3.88/5), while MentalEval achieves strong expert alignment with an average QWK of 0.944. Benchmarking shows that even the strongest LLM trails clinicians by 37.28 percentage points in objective psychiatric competence, with mental status assessment as a key bottleneck. Code and data are available at Anonymous GitHub.

1 Introduction

Large language models (LLMs) have made substantial progress in psychiatry [1, 2]. Existing studies have reported strong performance on key clinical components, including dialogue [3–6], medical record understanding [7], diagnostic reasoning [8], and treatment planning [9]. However, success on isolated clinical components remains insufficient to reflect the integrated capability required in real-world clinical encounters.

Recent studies have increasingly recognized the limitations of isolated evaluation [10, 11]. Accordingly, several works have begun exploring virtual medical environments for more comprehensive clinical simulation and assessment [12–15]. Yet, as summarized in Table 1, these environments still exhibit three limitations: *(1) Encounter Incompleteness*. Existing environments still center on doctor–patient communication, rather than complete encounters across multiple clinical scenarios and tasks. *(2) Data Inauthenticity*. The environmental evidence and reference targets are often derived from synthetic or web-based data, limiting clinical fidelity. *(3) Patient Simplification*. Virtual patients are often reactive symptom carriers, rather than faithful and distinctive psychiatric individuals.

To address these limitations, we introduce **MentalHospital**, to our knowledge the first virtual evaluation environment designed for psychiatric clinical encounters. MentalHospital moves beyond dialogue-centered simulation by instantiating the full S.O.A.P. workflow [16], from interviewing and examination to diagnosis and treatment planning. It is grounded in 1,193 de-identified psychiatric

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Table 1: Comparison of psychiatric benchmarks. *Inter.*, *Exam.*, *Diag.*, and *Treat.* denote Interview, Examination, Diagnosis, and Treatment Planning, respectively. MentalHospital provides EHR-grounded full-process simulation, multi-perspective evaluation, and specialized evaluators.

Benchmark	Data		Simulation		Evaluation		
	Data Source	Psychiatry	Scope	Interaction	Method	Reference	Evaluator
HealthBench [3]	Physician-authored	✗	General Healthcare	✗	Rubric	Physician	✗
MedDialogRubrics [17]	Synthetic	✗	Consultation / Diagnosis	✗	Rubric	✗	✗
CPsyCoun [4]	Web	✓	Consultation	✗	Rubric / Metrics	✗	✗
LingxiDiagBench [18]	Synthetic	✓	Consultation / Diagnosis	✗	Labels	✗	✗
PSYCHE [19]	Synthetic	✓	Consultation / Diagnosis	✓	Metrics	✗	✗
AI Hospital [13]	Web	✗	Consultation / Diagnosis	✓	Metrics	✗	✗
OpenHospital [14]	Synthetic	✗	Interview / Diagnosis	✓	Metrics	✗	✗
AgentClinic [12]	Single Hospital	✗	Inter. / Exam. / Diag.	✓	Metrics / Scale	EHR	✗
MentalHospital (Ours)	**Clinical Centers**	✓	**Inter. / Exam. / Diag. / Treat.**	✓	**Rub. / Metr. / Lab. / Scale**	**EHR / Psychiatrist**	**MentalEval**

EHR cases covering all major ICD-11 psychiatric categories and 76 disorder-level diagnoses, and constructs skill-augmented standardized patients with individualized symptom presentation and dynamic memory recall. In each interaction, LLMs must elicit evidence from the standardized patient, request auxiliary examinations from hospital-side modules, generate clinical notes and disorder-level decisions, and finally produce individualized treatment plans.

MentalHospital evaluates each encounter through *objective comparison* against EHR-derived references and *subjective assessment* of clinical process quality. The former can be computed from structured clinical targets, whereas the latter requires specialist judgment and is difficult to scale with clinician review alone. Our validation further shows that general-purpose LLM judges are insufficiently reliable for specialist clinical assessment: LLM-as-a-Judge achieves only a QWK of 0.677, an accuracy of 0.225, and an MAE of 0.875. To address this gap, we develop **MentalEval**, a suite of five domain-specific evaluators covering communication empathy, interviewing professionalism, clinical-note quality, diagnostic rigor, and treatment appropriateness. MentalEval is trained in two stages: *(1) Rubric-grounded SFT*, which teaches evaluators to generate rubric-based rationales and ratings from multi-level evaluation trajectories, and *(2) Expert-guided DPO*, which further aligns evaluator preferences with clinician-selected judgments.

We conduct experiments with LLMs, clinicians, medical trainees, and crowdworkers. Expert and trainee ratings validate the clinical fidelity of MentalHospital, with an overall score of 3.96/5, while ablations show that the full patient construction achieves the best objective faithfulness and subjective fidelity. MentalEval also achieves strong expert alignment, reaching an average QWK of 0.944 after DPO and outperforming LLM-as-a-Judge, Multi-LLM Jury, and individual crowdworker baselines. Using this evaluation setting, we find that current LLMs still lag behind clinicians in objective psychiatric competence: even the strongest medical-specific model trails medical trainees by 27.17 percentage points and human experts by 37.28 points on average across objective metrics, while showing complementary potential in empathic communication and treatment assistance. Fine-grained analysis further reveals a key weakness in mental status assessment. Finally, survey responses from 22 clinicians indicate positive acceptance of MentalHospital and its perceived value for psychiatric training. Our contributions are summarized as follows:

- We introduce **MentalHospital**, an EHR-grounded benchmark environment for full S.O.A.P. psychiatric encounters, using skill-augmented standardized patients constructed from 1,193 de-identified psychiatric EHR cases spanning all major ICD-11 categories and 76 disorders.
- We design a dual-track evaluation protocol that combines objective comparison against EHR-derived references with subjective assessment of clinical process quality, and develop **MentalEval**, a suite of specialist-aligned evaluators for scalable subjective assessment.
- We conduct comprehensive experiments with LLMs, clinicians, medical trainees, and crowdworkers, validating MentalHospital's clinical fidelity and MentalEval's expert alignment while revealing persistent gaps between current LLMs and clinicians in mental status assessment.

2 MentalHospital

2.1 Data Processing

MentalHospital is constructed from 1,193 de-identified psychiatric EHR cases obtained through an approved collaborative data-use partnership with a coordinating medical center. The resulting dataset comprises multi-center records sourced from six medical centers across three regions. The dataset includes 885 single-diagnosis cases and 308 comorbid cases. These cases cover all major ICD-11

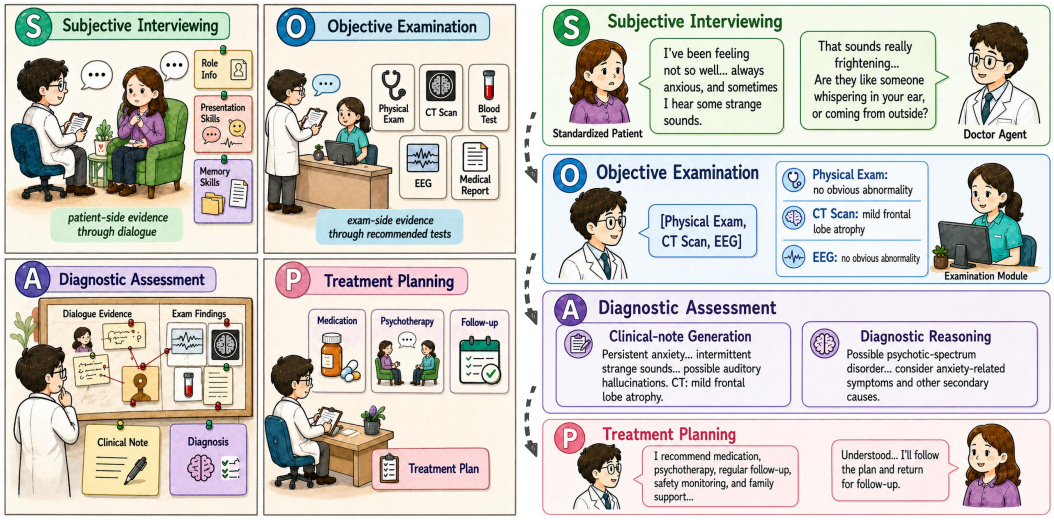


Figure 1: Overview of MentalHospital. A virtual psychiatric evaluation environment for LLM-based medical AI, built on the S.O.A.P. workflow and designed for dual-track evaluation.

psychiatric categories and 76 disorder-level diagnoses, providing broad disorder-level diversity for virtual psychiatric simulation (see Appendix F). Each case was independently reviewed by two licensed psychiatrists to ensure clinical completeness and consistency (Cohen's $\kappa > 0.8$). Details see Appendix E. Each EHR case is converted into a structured MentalHospital case c with patient-side evidence, examination-side evidence, and reference clinical targets:

$$x_c = \{\mathcal{K}_c^{\text{pat}}, \mathcal{K}_c^{\text{exam}}\}, \qquad \mathcal{Y}_c^* = \{y_c^{\text{cat}*}, y_c^{\text{dis}*}, y_c^{\text{treat}*}\}. \tag{1}$$

Here, $\mathcal{K}_c^{\text{pat}}$ contains patient-side evidence such as chief complaint, illness history, past and family history, and mental status; $\mathcal{K}_c^{\text{exam}}$ contains examination-side evidence such as physical and auxiliary examination findings. The reference targets $\mathcal{Y}_c^*$ include the diagnostic category $y_c^{\text{cat}*}$, the disorder-level diagnosis $y_c^{\text{dis}*}$, and EHR-derived treatment information $y_c^{\text{treat}*}$. In MentalHospital, $\mathcal{K}_c^{\text{pat}}$ is used to instantiate the standardized patient, $\mathcal{K}_c^{\text{exam}}$ supports the hospital-side examination module, and $\mathcal{Y}_c^*$ serves as the reference for diagnostic and treatment evaluation. Details see Appendix F.

To protect privacy while preserving clinical meaning, each EHR is processed using an on-premise `DeepSeek-R1` [20], rule-based detectors, and expert review. Sensitive events, exact dates, locations, institutions, and other identifiable details are generalized or masked. Clinically relevant sensitive events are rewritten into semantically similar but non-identifiable descriptions, so that psychiatric evidence remains usable for simulation and evaluation.

The study protocol and annotation procedures were approved by the institutional ethics boards of all participating clinical centers. All data collection, processing, and annotation followed the *Declaration of Helsinki*. Since records were de-identified before construction, individual informed consent was waived. All processing was conducted within secure hospital environments using locally deployed models, and no identifiable patient information or intermediate records left institutional governance.

2.2 Interaction Process

Given a psychiatric case c, MentalHospital instantiates a case-conditioned sequential environment:

$$\mathcal{E}_c = \{\mathcal{P}_c, \mathcal{M}_c^{\text{exam}}\}, \tag{2}$$

where $\mathcal{P}_c$ denotes the standardized patient constructed from patient-side evidence $\mathcal{K}_c^{\text{pat}}$, and $\mathcal{M}_c^{\text{exam}}$ denotes the hospital-side examination module supported by examination-side evidence $\mathcal{K}_c^{\text{exam}}$. The doctor agent cannot directly access $\mathcal{K}_c^{\text{pat}}$, $\mathcal{K}_c^{\text{exam}}$, or $\mathcal{Y}_c^*$, but must recover clinical evidence and infer decisions through interaction.

As shown in Figure 1, MentalHospital follows the clinical S.O.A.P. workflow. In the subjective interviewing stage, the doctor asks psychiatric interview questions and receives patient responses from $\mathcal{P}_c$, aiming to recover patient-side evidence such as symptoms, illness history, and mental status. In the objective examination stage, the doctor recommends examinations, and $\mathcal{M}_c^{\text{exam}}$ returns the

corresponding EHR-derived results when available. In the diagnostic assessment stage, the doctor integrates the dialogue trajectory and examination findings to generate a clinical note and diagnostic hypotheses. In the treatment planning stage, the doctor produces a treatment plan based on the accumulated evidence and diagnostic assessment. The interaction history is updated step by step:

$$h_t = h_{t-1} \cup \{(a_t, o_t)\}, \tag{3}$$

where a_t is the doctor action and o_t is the environment observation returned by either the standardized patient or the examination module. After completing the workflow, each episode produces

$$\hat{y}_c = \left\{ \tau_c,\, \hat{\mathcal{A}}_c^{\text{exam}},\, \hat{n}_c,\, \hat{d}_c,\, \hat{p}_c \right\}, \tag{4}$$

where τ_c is the doctor–patient trajectory, $\hat{\mathcal{A}}_c^{\text{exam}}$ is the recommended examination set, $\hat{n}_c$ is the clinical note, $\hat{d}_c$ is the diagnosis with reasoning, and $\hat{p}_c$ is the treatment plan. Detailed interaction details are provided in the Appendix G.

2.3 Environment Modules

MentalHospital consists of an environment controller, a standardized patient agent, a hospital-side examination module, and the doctor agent under evaluation. The controller manages the S.O.A.P. stage transitions, routes doctor actions to the corresponding module, records intermediate outputs, and issues instructions for completing the episode. See the Appendix H for implementation details.

Environment Controller. The controller coordinates the S.O.A.P. interaction process. It maintains the interaction history h_t, records the doctor–patient trajectory τ_c, collects examination requests $\hat{\mathcal{A}}_c^{\text{exam}}$, and stores downstream outputs including the clinical note $\hat{n}_c$, diagnosis $\hat{d}_c$, and treatment plan $\hat{p}_c$. It ensures that the doctor agent interacts with the appropriate module at each stage.

Standardized Patient. The standardized patient $\mathcal{P}_c$ is constructed from patient-side evidence $\mathcal{K}_c^{\text{pat}}$ as a skill-augmented patient configuration:

$$\mathcal{C}_c = \{\mathcal{R}_c, \mathcal{S}_c^{\text{pre}}, \mathcal{S}_c^{\text{mem}}\}, \tag{5}$$

where $\mathcal{R}_c$ specifies role information, $\mathcal{S}_c^{\text{pre}}$ encodes individualized symptom presentation grounded in the EHR mental status examination, and $\mathcal{S}_c^{\text{mem}}$ stores patient-side evidence as topic-level memory units. During interaction, the patient maintains an active memory state rather than exposing all memory units at once. Given the doctor action a_t and history h_{t-1}, the memory skill determines whether the current topic should be maintained or switched:

$$m_t = \text{UpdateMem}(m_{t-1}, a_t, h_{t-1}, \mathcal{S}_c^{\text{mem}}). \tag{6}$$

After updating the active memory, the patient generates a content-level response from the current recalled experience and role information, and then renders it with the presentation skill:

$$\tilde{u}_t^{\text{pat}} \sim \pi_{\text{pat}}^c(\cdot \mid h_{t-1}, a_t, \mathcal{R}_c, m_t), \qquad o_t^{\text{pat}} = \text{Render}(\tilde{u}_t^{\text{pat}}, \mathcal{S}_c^{\text{pre}}). \tag{7}$$

The final observation o_t^{pat} contains both behavioral presentation and verbal response, enabling the doctor agent to infer clinical evidence from both how the patient presents and what the patient states. This hybrid design ensures that the patient agent discloses only information supported by the retrieved memory while avoiding unrelated, unasked, or future-stage clinical information.

Examination Module. The examination module $\mathcal{M}_c^{\text{exam}}$ is implemented as a hospital-side question-answering interface based on `DeepSeek-V4-Flash` [20]. It receives examination requests from the doctor agent and returns results according to the examination-side evidence $\mathcal{K}_c^{\text{exam}}$. If a requested examination exists in $\mathcal{K}_c^{\text{exam}}$, the module returns the corresponding EHR-derived report; otherwise, it returns *"result unavailable"*. The module also records the requested examination set $\hat{\mathcal{A}}_c^{\text{exam}}$ for comparison with the EHR-derived reference set when calculating examination coverage.

2.4 Dual-track Evaluation

MentalHospital evaluates each completed episode through *objective comparison* and *subjective assessment*. Objective comparison measures outcome-level task correctness against EHR-derived references. Subjective assessment evaluates process-level clinical professionalism and interpretability using expert-defined rubrics. Detailed metrics are provided in Section 4.

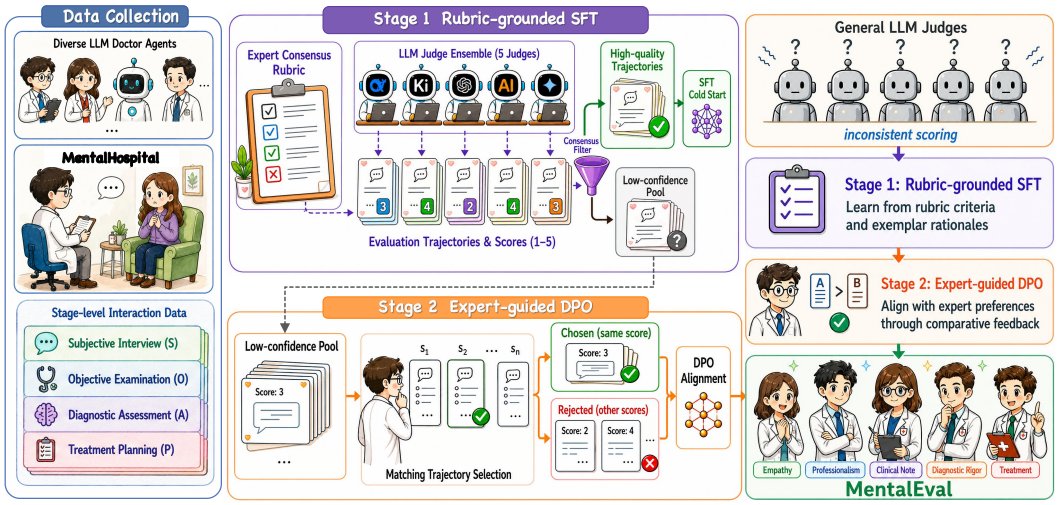


Figure 2: Overview of MentalEval construction. MentalEval is trained on MentalHospital interaction data via rubric-grounded SFT and expert-guided DPO for subjective clinical evaluation.

3 MentalEval

We explored two approaches to scalable subjective evaluation: crowdworker review and rubric-prompted general-purpose LLM judges. Experiment 4.4 shows that crowdworker review is reasonably aligned with specialists (QWK = 0.904, Acc. = 0.625, MAE = 0.375), yet costly to scale, while LLM-as-a-Judge shows limited specialist agreement (QWK = 0.677, Acc. = 0.225, MAE = 0.875).We therefore develop **MentalEval**, a suite of five Qwen3-8B-based domain-specific evaluators trained with rubric-grounded SFT followed by expert-guided DPO. The evaluators assess communication empathy, interviewing professionalism, clinical-note quality, diagnostic rigor, and treatment appropriateness. Given an evaluation context such as dialogue, clinical note, diagnostic reasoning, or treatment recommendation, each evaluator outputs a rubric-grounded rationale and a Likert-style 1–5 rating. Detailed rubrics are provided in Appendix I.

3.1 Stage 1: Rubric-grounded SFT

The first stage provides a cold start for learning rubric-grounded evaluation patterns. We sample 200 cases for evaluator training, then select 20 LLMs covering different providers, model scales, and specialties, and run each model three times on each case, yielding complete interaction episodes. As shown in Figure 2, these episodes are further decomposed into stage-level interaction data for the five subjective evaluation dimensions. For each stage-level sample z_i, we use a judge ensemble $\mathcal{J}$ of five LLMs (`DeepSeek-V4-Pro` [20], `KIMI-K2.6` [21], `GPT-5.5` [22], `Claude-Opus-4.6` [23], and `Gemini-3.1-Pro` [24]) to generate rubric-based evaluation trajectories and ratings:

$$(s_{i,j}, r_{i,j}) = J_j(z_i, \mathcal{B}), \qquad j \in \mathcal{J}, \quad r_{i,j} \in \{1, \ldots, 5\}, \tag{8}$$

where $\mathcal{B}$ denotes the expert consensus rubric, $s_{i,j}$ is the generated evaluation trajectory, and $r_{i,j}$ is the assigned score. To filter reliable supervision, we retain only judge outputs with sufficient score agreement. Specifically, the consensus score for z_i is defined as:

$$r_i^* = \arg \max_{r \in \{1,\ldots,5\}} \sum_{j \in \mathcal{J}} \mathbf{1}[r_{i,j} = r]. \tag{9}$$

If at least k judges assign r_i^*, the trajectories with score r_i^* are retained as high-quality SFT targets; otherwise, the candidate set is treated as low-confidence and reserved for preference construction. Because extreme scores are underrepresented, we additionally select near-boundary samples and rewrite 2-point cases toward 1-point quality and 4-point cases toward 5-point quality. These augmented samples are retained only after judge-ensemble verification. The SFT objective is:

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E}_{(z,s)\sim\mathcal{D}_{\text{SFT}}} \log \pi_\theta(s \mid z). \tag{10}$$

3.2 Stage 2: Expert-guided DPO

After cold-start training, we further align MentalEval with specialist judgment using expert-guided preference optimization. For each low-confidence candidate set $\mathcal{S}_i = \{(s_{i,1}, r_{i,1}), \ldots, (s_{i,5}, r_{i,5})\}$,

Table 2: Representative objective and subjective evaluation results across S.O.A.P. stages. Objective metrics are reported as percentages (%), while subjective scores are reported on a 1–5 scale.

Model	Objective Comparison (%)					Subjective Assessment (1–5)				
	Inter.	Exam.	Note	Cate.	Diso.	Emp.	Prof.	Qual.	Rigor	App.
Medical Trainees	78.23	71.05	82.91	81.42	72.58	1.63	3.86	3.87	2.33	2.47
Human Experts	82.35	91.90	88.67	90.59	83.24	1.22	4.25	4.01	2.54	2.96
Small Language Models										
Llama-3-8B	19.06	7.94	9.96	18.11	6.08	2.05	2.88	1.93	1.86	2.67
Qwen3-14B	23.19	8.33	12.12	23.55	15.19	1.96	2.90	2.17	2.20	2.83
GPT-OSS-20B	27.43	8.60	33.12	37.21	25.81	1.60	3.15	3.14	2.06	3.23
QWQ-32B	30.40	10.87	32.30	44.53	26.39	2.13	3.72	2.85	3.14	3.40
Large Language Models										
GPT-OSS-120B	36.72	9.92	39.94	49.54	34.19	1.50	3.45	3.89	2.42	3.35
Claude-Sonnet-4.6	30.44	21.28	34.81	50.28	34.57	1.72	3.39	2.70	2.22	3.17
Deepseek-v4-Pro	30.95	21.93	56.46	26.20	20.85	2.09	3.48	2.91	2.39	3.29
GPT-5.5	69.06	22.99	45.44	70.23	49.85	1.50	3.63	3.66	2.21	3.48
Medical-specific Models										
MedGemma-4B-it	25.65	9.42	10.67	21.48	3.05	2.16	2.57	1.99	2.15	2.91
MedGemma-27B-it	43.12	16.05	35.91	32.27	10.10	2.05	2.83	2.67	2.33	3.17
Baichuan-M2	51.29	24.76	50.54	80.56	43.19	1.29	2.85	2.91	2.44	3.33

we present the original stage-level interaction data z_i and the candidate evaluation trajectories to clinicians. The clinician selects the trajectory s_i^* whose rating and rationale best match the clinical quality of z_i. Candidate trajectories with the same score as s_i^* are treated as chosen responses, while the remaining trajectories are treated as rejected responses:

$$\mathcal{S}_i^+ = \{s_{i,j} \mid r_{i,j} = r_i^*\}, \qquad \mathcal{S}_i^- = \{s_{i,j} \mid r_{i,j} \neq r_i^*\}. \tag{11}$$

We then construct preference triples (z_i, s_i^+, s_i^-) for DPO training. The DPO objective is:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(z,s^+,s^-)} \log \sigma \left(\beta \left[\log \frac{\pi_\theta(s^+ \mid z)}{\pi_{\text{ref}}(s^+ \mid z)} - \log \frac{\pi_\theta(s^- \mid z)}{\pi_{\text{ref}}(s^- \mid z)} \right] \right). \tag{12}$$

Detailed MentalEval training settings are provided in Appendix J.

4 Experiments

Task Definition. We formulate evaluation as a multi-stage, mixed-type clinical task. Given a case c, a doctor agent completes an episode covering subjective interviewing, objective examination, diagnostic assessment, and treatment planning. The task combines three forms: (1) open-ended generation, including dialogue, clinical-note writing, diagnostic reasoning, and treatment planning; (2) multi-label decision making, including examination recommendation and category-level diagnosis; and (3) ranked disorder prediction, for definitive diagnosis. We evaluate the resulting trajectory and stage outputs from two perspectives: (i) objective comparison, which measures EHR-grounded outcomes, and (ii) subjective assessment, which measures process-level clinical quality.

Evaluation Metrics. For EHR-grounded outputs, including examination recommendation (Exam.), clinical interviewing (Inter.), and clinical-note generation (Note), we measure reference coverage:

$$\text{Coverage}(y_c, \mathcal{K}_c) = \frac{1}{|\mathcal{K}_c|} \sum_{k \in \mathcal{K}_c} \mathbb{I}\left[k \preceq y_c\right], \tag{13}$$

where $\mathcal{K}_c$ denotes the EHR-derived reference checkpoint set for case c, and $k \preceq y_c$ indicates that checkpoint k is semantically entailed by the generated output y_c. We use a conservative semantic similarity threshold of $\tau = 0.85$ for high-precision automatic matching, and adjudicate pairs below the threshold with an empirically validated Multi-LLM-assisted semantic matching procedure. For definitive diagnosis, we report category identification (Cate.) and disorder-level diagnosis ranking (Diso.), with nDCG used to evaluate diagnostic prioritization. For subjective assessment, MentalEval scores communication empathy (Emp.), interviewing professionalism (Prof.), clinical-note quality (Qual.), diagnostic rigor (Rigor), and treatment appropriateness (App.) using expert-defined rubrics. For more details see Appendix K.

Table 4: Ablation of skill-augmented patient construction for standardized patients.

Patient Variant	Objective Comparison (%)			Subjective Assessment (1–5)		
	Cov.↑	Prec.↑	Halluc.↓	Realism↑	Fidelity↑	Distinct.↑
Naive Prompting	72.40	86.10	7.80	2.12	2.23	1.92
Representation Skill Only	$72.40_{+0.00}$	$85.72_{-0.38}$	$9.04_{+1.24}$	$4.12_{+2.00}$	$3.82_{+1.59}$	$3.72_{+1.80}$
Memory Skill Only	$84.83_{+12.43}$	$91.64_{+5.92}$	$4.52_{-4.52}$	$3.55_{-0.57}$	$3.87_{+0.05}$	$2.77_{-0.95}$
Full Patient	$88.91_{+4.08}$	$93.27_{+1.63}$	$3.08_{-1.44}$	$4.17_{+0.62}$	$4.05_{+0.18}$	$3.87_{+1.10}$

Comparison Groups. We consider five comparison groups: human experts, medical trainees, crowdworkers, general-purpose LLMs, and medical-specific LLMs. The expert group consists of 5 licensed psychiatrists and 3 psychology experts. We also include 14 medical trainees and 8 non-specialist crowdworkers. For LLMs, we evaluate a set of general-purpose models, including Llama-3-8B [25], Qwen3-14B [26], GPT-OSS-20B [27], QwQ-32B [26], GPT-OSS-120B [27], DeepSeek-V4-Pro [20], GPT-5.5 [22], and Claude-Sonnet-4-6 [23]. We also include medical-oriented LLMs, including MedGemma-4B-it [28], MedGemma-27B-it [28], and Baichuan-M2 [29].

4.1 Can LLMs Interact Like Clinicians?

We evaluate 12 LLMs, 8 human experts, and 14 medical trainees in MentalHospital. As shown in Table 2, current LLMs still show a clear gap from clinicians in objective clinical competence. Compared with the strongest medical-specific model, Baichuan-M2, medical trainees achieve higher scores across all five objective metrics, with an average gap of 27.17 percentage points; human experts further widen this gap to 37.28 points. In subjective assessment, LLMs and clinicians show complementary strengths rather than uniform superiority of either group. Human experts exhibit stronger interviewing professionalism (4.25), reflecting more efficient, goal-directed clinical information gathering, but receive lower scores in expressed communication empathy (1.22). In contrast, the best-performing LLMs reach or exceed human experts in expressed empathy (2.16 vs. 1.22), diagnostic rigor (3.14 vs. 2.54), and treatment appropriateness (3.48 vs. 2.96), while remaining lower in interviewing professionalism (3.72 vs. 4.25) and clinical-note quality (3.89 vs. 4.01). These results suggest that LLMs are not yet reliable substitutes for clinicians, but may provide complementary value in verbal affective support and evidence-informed treatment assistance.

4.2 Where Do LLMs Fall Short in Clinical Interaction?

To identify the core gap, we decompose coverage in clinical interviewing and note generation into Chief Complaint (CC) and Mental Status (MS) in Table 3, current LLMs fail to recognize the clinical importance of mental status examination in psychiatric assessment and diagnosis. For example, `GPT-5.5` shows a large gap in interviewing (85.38 vs. 55.72), and `DeepSeek-v4-Pro` exhibits a similar pattern in note generation (59.94 vs. 43.91).

Table 3: Coverage by clinical component.

Model	Interview Phase			Note Phase		
	Avg.	CC	MS	Avg.	CC	MS
Qwen3-14B	23.19	44.36	12.57	12.12	16.38	17.25
MedGemma-4B-it	25.65	40.69	10.45	10.67	9.68	6.66
QWQ-32B	30.40	50.90	17.84	32.30	42.91	34.22
Deepseek-v4-Pro	30.95	41.67	23.13	56.46	59.94	43.91
GPT-5.5	69.06	85.38	55.72	45.44	55.34	65.89

4.3 Does Skill-Augmented Patient Construction Improve Fidelity?

We evaluate patient construction fidelity through objective evaluation and subjective assessment, with detailed experimental settings provided in Appendix L. As shown in Table 4, naive prompting achieves reasonable factual faithfulness but performs poorly in subjective fidelity, indicating that directly exposing the EHR does not yield realistic patient simulation. Representation skill substantially improves clinical realism and patient distinctiveness, while memory skill improves evidence coverage, precision, and hallucination control. The full construction achieves the best overall performance.

4.4 Does MentalEval Align with Expert Judgment?

MentalEval is designed as a task-specific evaluator for MentalHospital, rather than a general-purpose LLM judge across arbitrary clinical tasks. Our goal is therefore to test whether it can reproduce expert judgments within the MentalHospital evaluation setting.To avoid evaluation leakage, we additionally sampled 40 psychiatric EHR cases from the coordinating medical center to construct the MentalEval

Table 5: Ablation results of MentalEval training against expert ratings. DPO is applied after SFT.

Evaluator	Base Model (Qwen3-8B)			+SFT			+DPO		
	QWK↑	Acc.↑	MAE↓	QWK↑	Acc.↑	MAE↓	QWK↑	Acc.↑	MAE↓
MentalEval-Emp	0.677	0.225	0.875	$0.824_{+0.148}$	$0.500_{+0.275}$	$0.525_{-0.350}$	$0.926_{+0.102}$	$0.725_{+0.225}$	$0.275_{-0.250}$
MentalEval-Pro	0.705	0.275	0.825	$0.844_{+0.139}$	$0.550_{+0.275}$	$0.475_{-0.350}$	$0.941_{+0.097}$	$0.775_{+0.225}$	$0.225_{-0.250}$
MentalEval-Note	0.766	0.400	0.675	$0.910_{+0.144}$	$0.700_{+0.300}$	$0.300_{-0.375}$	$0.940_{+0.030}$	$0.775_{+0.075}$	$0.225_{-0.075}$
MentalEval-Diag	0.828	0.475	0.575	$0.887_{+0.059}$	$0.725_{+0.250}$	$0.325_{-0.250}$	$0.967_{+0.080}$	$0.875_{+0.150}$	$0.125_{-0.200}$
MentalEval-Treat	0.819	0.425	0.600	$0.928_{+0.109}$	$0.725_{+0.300}$	$0.275_{-0.325}$	$0.948_{+0.020}$	$0.800_{+0.075}$	$0.200_{-0.075}$
Average	0.759	0.360	0.710	$0.879_{+0.120}$	$0.640_{+0.280}$	$0.380_{-0.330}$	$0.944_{+0.066}$	$0.790_{+0.150}$	$0.210_{-0.170}$

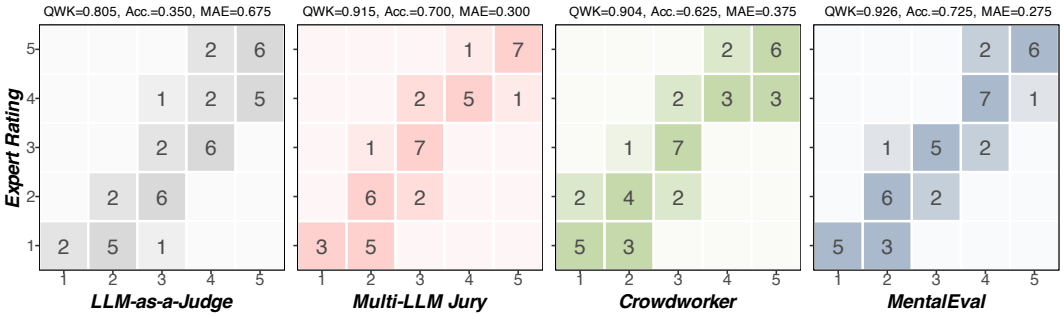


Figure 3: Agreement with expert ratings, where MentalEval shows the strongest alignment.

test set. These cases were strictly excluded from all MentalEval training data. For each case, we sample model-generated responses and evaluate them across the five MentalEval dimensions, yielding 200 dimension-level test samples in total. Each sample is independently rated by three experts, and only unanimously assigned samples are retained as expert-validated references. We ablate the two-stage training procedure on this held-out set of unseen expert-labeled cases. As shown in Table 5, rubric-grounded SFT consistently improves all five evaluators, increasing average QWK from 0.759 to 0.879 and reducing MAE from 0.710 to 0.380. Expert-guided DPO further improves agreement, reaching an average QWK of 0.944, accuracy of 0.790, and MAE of 0.210. We then compare the trained MentalEval with LLM-as-a-Judge, Multi-LLM Jury, and individual crowdworker ratings in Figure 3. MentalEval achieves the strongest expert alignment, with the highest QWK (0.926) and accuracy (0.725), and the lowest MAE (0.275).

4.5 Does MentalHospital Provide a Clinically Faithful Environment?

We conduct a survey to assess the clinical faithfulness with 5 psychiatrists, 3 psychologists, and 14 medical trainees (full questionnaire is see Appendix M). The questionnaire covers five dimensions: clinical realism (CR), patient fidelity (PF), patient distinctiveness (PD), evaluation credibility (EC), and training suitability (TS), using a 5-point Likert scale. As shown in Figure 4, MentalHospital receives consistently positive ratings, with an overall mean score of 3.88/5 (95% bootstrap CI: [3.78, 3.98]), a positive response rate of 69.9% (95% CI: [64.8%, 75.0%]), and a negative response rate of only 5.1% (95% CI: [3.4%, 6.8%]). The ratings are particularly strong for training suitability and evaluation credibility, with positive response rates of 84.8% and 75.0%, suggesting that MentalHospital provides a clinically plausible environment.

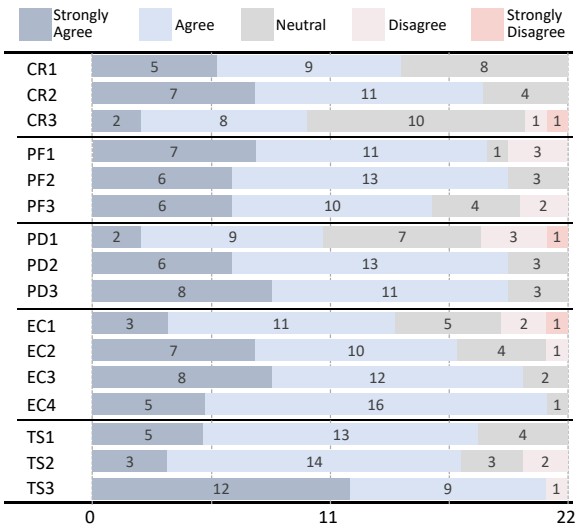


Figure 4: Clinical fidelity ratings for MentalHospital.

4.6 Qualitative Error Analysis

Figure 5 illustrates two representative failure modes observed in MentalHospital. In a case involving anxiety and possible perceptual disturbance, `GPT-5.5` directly probes hallucination-related features such as source and content, yielding higher objective evidence coverage than `Qwen3-14B`. However, its response contains limited empathic acknowledgment, resulting in a lower MentalEval empathy score. Conversely, `Qwen3-14B` provides more explicit emotional validation and reassurance, but fails to pursue the hallucination hypothesis through targeted psychiatric questioning. This contrast shows that evidence-seeking behavior and process-level communication quality can diverge substantially. A model may recover key diagnostic evidence while offering poor affective support, whereas another may appear empathic but remain clinically under-informative.

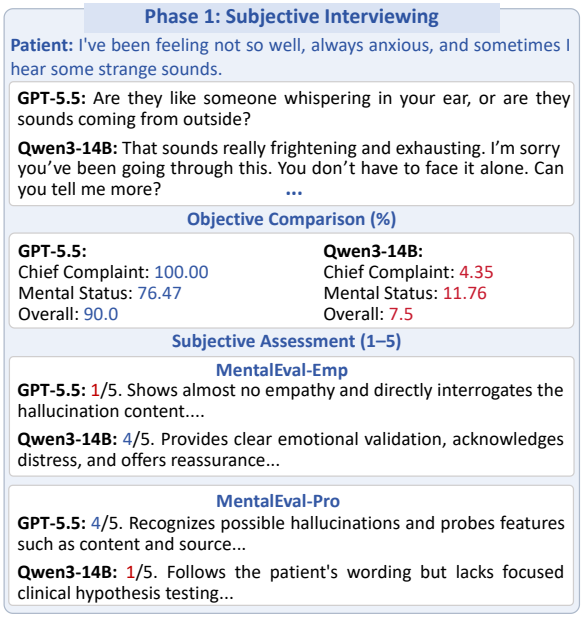


Figure 5: Representative cases in MentalHospital.

5 Related Work

Psychiatric Benchmarks. Early medical LLM evaluation mainly relied on static question-answering benchmarks, testing medical knowledge recall and exam-style clinical reasoning [30, 31]. This paradigm was later expanded to broader medical evaluation suites and real-world clinical benchmarks covering diverse tasks, safety, and clinical reliability [32–34, 3], with subsequent work further examining clinical safety, expert consensus, and specialized reasoning [35–38]. In parallel, mental-health benchmarks began to assess counseling, emotional support, and psychotherapy [11, 39, 9], while clinically oriented psychiatric benchmarks focused on psychiatric practice, disorder recognition, and diagnostic reasoning [40, 41, 8, 42]. These efforts connect subclinical mental-health support with threshold-level psychiatric assessment, but still largely evaluate isolated responses or task-specific outputs. Recent work has moved toward multi-turn dialogue and simulated clinical environments, testing contextual interaction across turns [4–6, 43–45] and patient–doctor simulation [12–14, 46, 47, 18]. However, existing benchmarks still rarely support virtual psychiatric encounters.

Evaluation Methods. Early evaluation of LLM outputs mainly relied on task accuracy and surface-level automatic metrics, including lexical overlap and embedding-based similarity [48–51]. These metrics remain common in benchmark-based assessment and text generation evaluation [52, 34], but they are insufficient for clinically nuanced behaviors such as empathy, professionalism, diagnostic rigor, and treatment appropriateness. Later work therefore incorporated clinician-centered or expert-based protocols [53, 3], with similar trends in clinical safety and reasoning benchmarks [35, 54]. Although expert review improves clinical validity, it is difficult to scale. Recent studies have explored model-based judging, including judge-style, jury-style, and rubric-driven frameworks [55, 33, 56], while also highlighting reliability concerns of LLM-based judges in medical settings [57–59]. Existing methods therefore still struggle to balance expertise and scalability, especially for subjective mental-health evaluation where process quality cannot be reduced to objective correctness alone.

6 Conclusion

We present MentalHospital, a virtual psychiatric evaluation environment for LLM-based medical AI. MentalHospital supports interactive S.O.A.P.-based psychiatric assessment with EHR-grounded virtual patients and a dual-track evaluation protocol that combines objective comparison with scalable subjective assessment. To operationalize specialist judgment, we develop MentalEval, a suite of five domain-specific evaluators trained through rubric-grounded SFT and expert-guided DPO. Experiments show that current LLMs remain limited in clinician-like psychiatric interaction, especially in mental status assessment, while MentalHospital and MentalEval provide faithful and scalable tools for evaluating in psychiatric care. Finally, clinician survey responses indicate positive acceptance of MentalHospital and its perceived value for psychiatric training.